



SYDNEY BOYS HIGH  
MOORE PARK, SURRY HILLS

**SEPTEMBER 2011**  
**YEARLY**  
**YEAR 11 Continuers**

# Mathematics Ext 1

## General Instructions:

- Reading time—5 minutes.
- Working time—90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Answer in simplest exact form unless otherwise stated.

## Total marks—68 Marks

- Attempt questions 1–4.
- The mark-value of each question is boxed in the right margin.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 2 sections:  
Section A (Questions 1 and 2),  
Section B (Questions 3 and 4).

**Examiner:** Mr P. Bigelow

## Section A

Marks

### Question 1 (17 marks)

(a) Differentiate the following:

5

(i)  $y = \frac{4}{3\sqrt{x}},$

(ii)  $y = \frac{x+1}{x-2},$

(iii)  $y = (2 - 3x^2)^5.$

(b) Find the equation of the normal to  $y = \frac{3}{x}$  at the point where  $x = -3.$

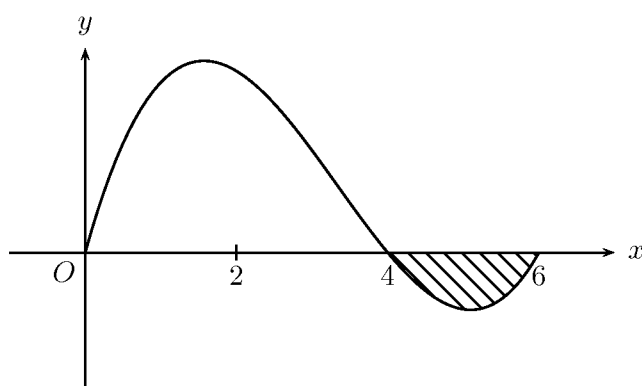
2

(c) Differentiate  $y = x^2 + bx + c$ , and hence find  $b$  and  $c$  if the parabola has a  $y$ -intercept of  $-3$  and its gradient at that point is  $-2.$

3

(d)

2



You are given that  $\int_0^4 f(x) dx = 22$  and  $\int_0^6 f(x) dx = 15.$

(i) What is the area of the shaded region?

(ii) What is the value of  $\int_4^6 f(x) dx$ ?

(e) Find  $\int_1^2 (2x - 3x^2) dx.$

2

(f)  $\theta$  and  $\phi$  are acute angles and  $\cos \phi = \frac{1}{3}$  and  $\sin \theta = \frac{2}{3}$ , find the exact value of  $\cos(\theta - \phi).$

3

**Question 2 (17 marks)**

(a) Consider the function  $f(x) = 2 + 3x - x^3$ .

7

(i) Find  $f'(x)$ .

(ii) Find  $f''(x)$ .

(iii) Locate any stationary points and determine their nature.

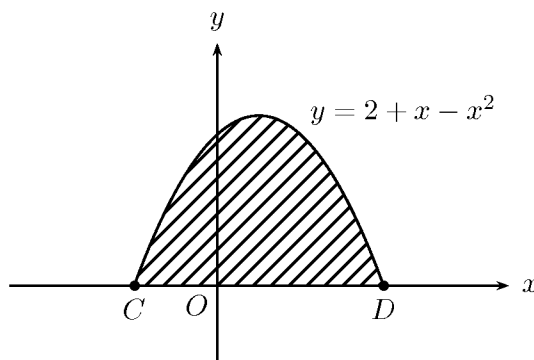
(iv) For what values of  $x$  is the curve increasing?

(v) For what values of  $x$  is the curve concave up?

(vi) Sketch the curve.

(b)

4



(i) Find the co-ordinates of  $C$  and  $D$ .

(ii) Find the ratio in which the shaded area is divided by the  $y$ -axis.

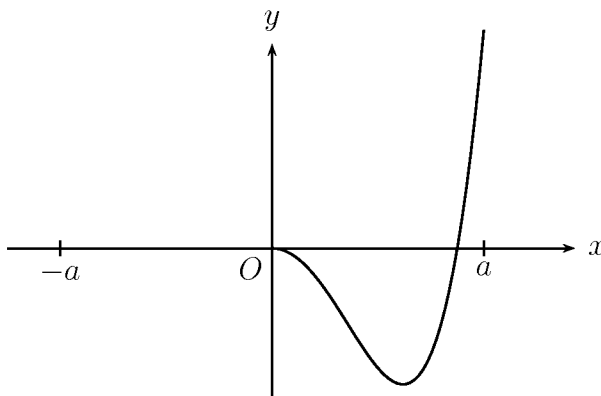
(c) (i) Expand  $\sin(A + B)$ .

3

(ii) Hence show that  $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$ .

(d)

3



The diagram shows the graph of the function for  $0 \leq x \leq a$ .

It is known that  $f(x)$  is an odd function and stationary at  $(0, 0)$ .

(i) Copy the diagram and continue the graph for  $-a \leq x < 0$ .

(ii) On another diagram, sketch  $y = f'(x)$ .

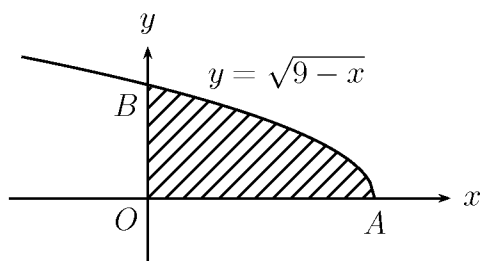
## Section B

(Use a separate writing booklet.)

Marks

### Question 3 (20 marks)

(a)



The shaded area is bounded by the curve  $y = \sqrt{9-x}$  and the co-ordinate axes.

7

(i) Write down the values of  $A$  and  $B$ .

(ii) Find the area  $AOB$ .

(iii) Find the volume when the shaded area is rotated about

( $\alpha$ ) the  $x$ -axis,

( $\beta$ ) the  $y$ -axis.

(b) Solve the following inequalities and graph the solutions on separate number lines.

4

(i)  $\frac{2}{|x-1|} > 1,$

(ii)  $\frac{3}{x-2} \leq \frac{2}{x+1}.$

(c) Given  $\triangle ABC$  with vertices  $A(3, 2)$ ,  $B(5, 6)$  and  $C(7, 4)$ :

5

(i) Find the co-ordinates of  $E$  and  $F$ , the midpoints of  $AB$  and  $BC$  respectively.

(ii) Show that  $EF$  is

( $\alpha$ ) half of

( $\beta$ ) and parallel to  $AC$ .

(d) Prove the identity

2

$$\frac{\sin 2A}{1 + \cos 2A} = \tan A.$$

(e) Solve  $4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 1$  for  $0 \leq \theta \leq 2\pi$ .

2

[HINT: use a double angle result.]

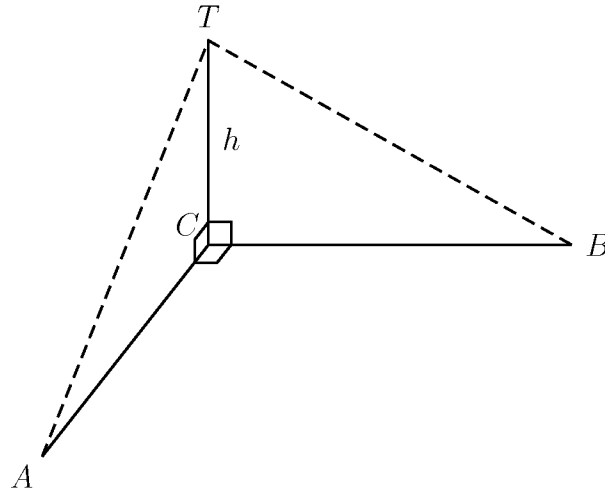
Question 4 (14 marks)

- (a) The graph of  $y = f(x)$  passes through  $(2, 1)$  and  $f'(x) = \frac{x^3 - 4}{x^2}$ . Find  $f(x)$ .

2

- (b)

4



$CT$  is a tower of height  $h$ . From  $A$ , due south of the base of the tower, the angle of elevation of  $T$ , the top of the tower, is  $25^\circ$ . From  $B$ , due east, the angle of elevation of  $T$  is  $30^\circ$ . Given that  $AB$  is 200 m, find  $h$  to the nearest metre.

- (c) Let  $f(x) = \frac{2x}{(1+x^2)^2}$ .

8

- (i) Copy this table and complete it in decimal form.

|      |   |   |   |
|------|---|---|---|
| x    | 1 | 2 | 3 |
| f(x) |   |   |   |

- (ii) Use the table to find values for  $\int_1^3 f(x) dx$  using  
 ( $\alpha$ ) the Trapezoidal rule,  
 ( $\beta$ ) Simpson's rule.

- (iii) Find  $\frac{d}{dx} \left( \frac{1}{x^2 + 1} \right)$ .

- (iv) Using (iii), find the exact value of the integral in (ii).

End of Paper

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$



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# **Mathematics Ext 1 Solutions**

## Section A

Marks

### Question 1 (17 marks)

(a) Differentiate the following:

5

(i)  $y = \frac{4}{3\sqrt{x}},$

**Solution:**  $y = \frac{4x^{-\frac{1}{2}}}{3}.$   
 $\frac{dy}{dx} = -\frac{1}{2} \cdot \frac{4x^{-\frac{3}{2}}}{3},$   
 $= -\frac{2}{3x\sqrt{x}}.$

(ii)  $y = \frac{x+1}{x-2},$

**Solution:**  $\frac{dy}{dx} = \frac{(x-2) \times 1 - (x+1) \times 1}{(x-2)^2},$   
 $= \frac{-3}{(x-2)^2}.$

(iii)  $y = (2 - 3x^2)^5.$

**Solution:**  $\frac{dy}{dx} = 5 \times (-3 \times 2x) \times (2 - 3x^2)^4,$   
 $= -30x(2 - 3x^2)^4.$

(b) Find the equation of the normal to  $y = \frac{3}{x}$  at the point where  $x = -3$ .

2

**Solution:**  $\frac{dy}{dx} = -\frac{3}{x^2},$   
 $= -\frac{1}{3}$  at the point  $(-3, -1).$

Normal equation:

$y + 1 = 3(x + 3),$   
*i.e.*,  $3x - y + 8 = 0.$

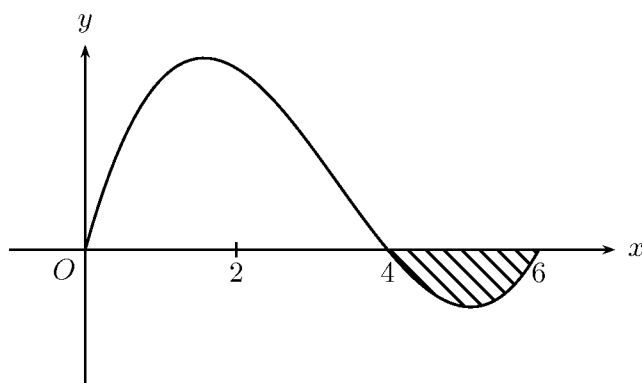
(c) Differentiate  $y = x^2 + bx + c$ , and hence find  $b$  and  $c$  if the parabola has a  $y$ -intercept of  $-3$  and its gradient at that point is  $-2$ .

3

**Solution:**  $y' = 2x + b.$   
 Now, at  $(0, -3)$  :  
 $-2 = 2 \times 0 + b,$   
*i.e.*,  $b = -2,$   
 and  $-3 = 0^2 - 2 \times 0 + c,$   
*i.e.*,  $c = -3.$



(d)



2

You are given that  $\int_0^4 f(x) dx = 22$  and  $\int_0^6 f(x) dx = 15$ .

(i) What is the area of the shaded region?

**Solution:** Shaded area =  $|15 - 22|$ ,  
 $= 7$ .

(ii) What is the value of  $\int_4^6 f(x) dx$ ?

**Solution:**  $\int_4^6 f(x) dx = \int_0^6 f(x) dx - \int_0^4 f(x) dx$ ,  
 $= -7$ .

(e) Find  $\int_1^2 (2x - 3x^2) dx$ .

2

**Solution:**  $\int_1^2 (2x - 3x^2) dx = \left[ x^2 - x^3 \right]_1^2$ ,  
 $= 4 - 8 - (1 - 1)$ ,  
 $= -4$ .

(f)  $\theta$  and  $\phi$  are acute angles and  $\cos \phi = \frac{1}{3}$  and  $\sin \theta = \frac{2}{3}$ , find the exact value of  $\cos(\theta - \phi)$ .

3

**Solution:**

$$\begin{aligned} \cos(\theta - \phi) &= \cos \theta \cos \phi + \sin \theta \sin \phi, \\ &= \frac{\sqrt{5}}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{2\sqrt{2}}{3}, \\ &= \frac{\sqrt{5} + 4\sqrt{2}}{9}. \end{aligned}$$

**Question 2 (17 marks)**(a) Consider the function  $f(x) = 2 + 3x - x^3$ .

7

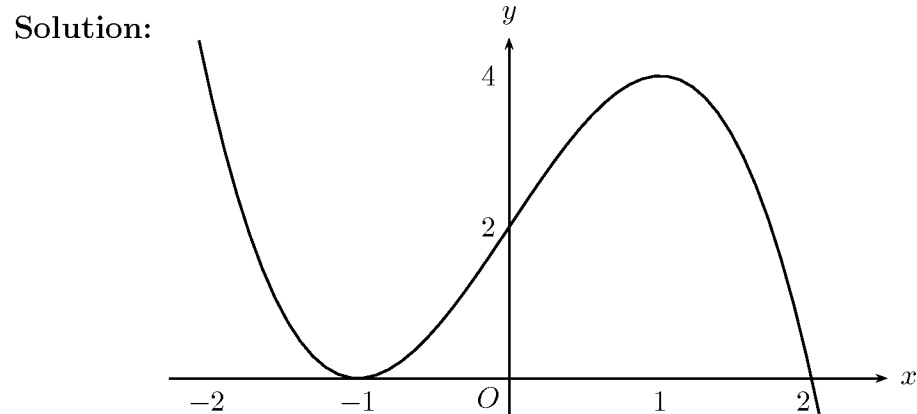
(i) Find  $f'(x)$ .**Solution:**  $f'(x) = 3 - 3x^2$ .(ii) Find  $f''(x)$ .**Solution:**  $f''(x) = -6x$ .

(iii) Locate any stationary points and determine their nature.

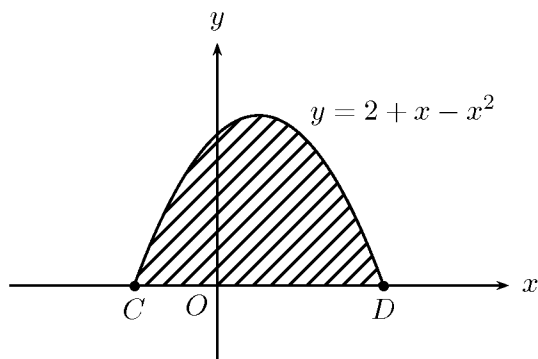
**Solution:** When  $f'(x) = 0$ , and when  $f''(x) = 0$ ,  
 $3(1 - x^2) = 0$ ,  $x = 0$ ,  $y = 2$ ,  
 $x = \pm 1$ ,  $f''(1) = -6$ ,  
 $y = 4, 0$ .  $f''(-1) = 6$ .  
 $\therefore$  A maximum at  $(1, 4)$  and a minimum at  $(-1, 0)$ .

(iv) For what values of  $x$  is the curve increasing?**Solution:**  $-1 < x < 1$ .(v) For what values of  $x$  is the curve concave up?**Solution:**  $x < 0$ .

(vi) Sketch the curve.



(b)

(i) Find the co-ordinates of  $C$  and  $D$ .

**Solution:**  $x^2 - x - 2 = 0,$   
 $(x - 2)(x + 1) = 0,$   
 $\therefore x = -1, 2.$   
*i.e.*,  $C(-1, 0)$  and  $D(2, 0).$

(ii) Find the ratio in which the shaded area is divided by the  $y$ -axis.

**Solution:**  $\int_{-1}^0 (2 + x - x^2) dx = \left[ 2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^0,$   
 $= 0 - \left\{ -2 + \frac{1}{2} - \frac{1}{3} \right\},$   
 $= 1\frac{5}{6}.$   
 $\int_0^2 (2 + x - x^2) dx = \left[ 2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_0^2,$   
 $= \left\{ 4 + \frac{4}{2} - \frac{8}{3} \right\} - 0,$   
 $= 5.$   
*i.e.*, the ratio is 11 : 30.

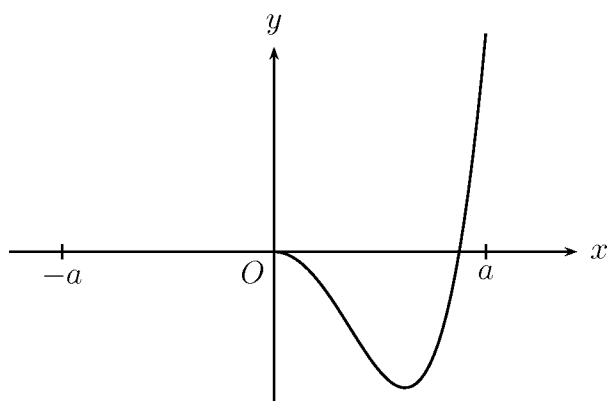
(c) (i) Expand  $\sin(A + B)$ .

**Solution:**  $\sin(A + B) = \sin A \cos B + \cos A \sin B.$

(ii) Hence show that  $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$ 

**Solution:**  $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ,$   
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2},$   
 $= \frac{\sqrt{6} + \sqrt{2}}{4}.$

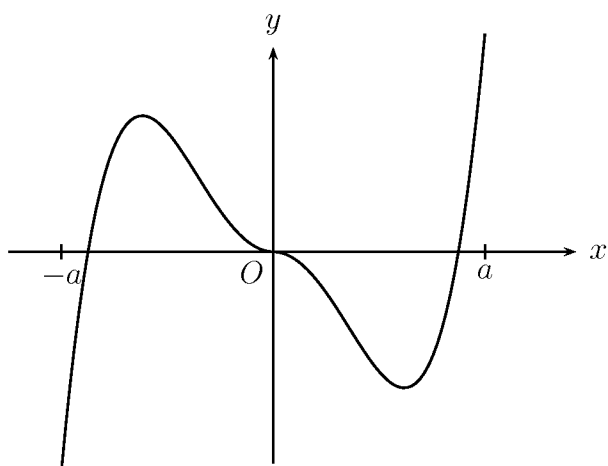
(d)



The diagram shows the graph of the function for  $0 \leq x \leq a$ .  
It is known that  $f(x)$  is an odd function and stationary at  $(0, 0)$ .

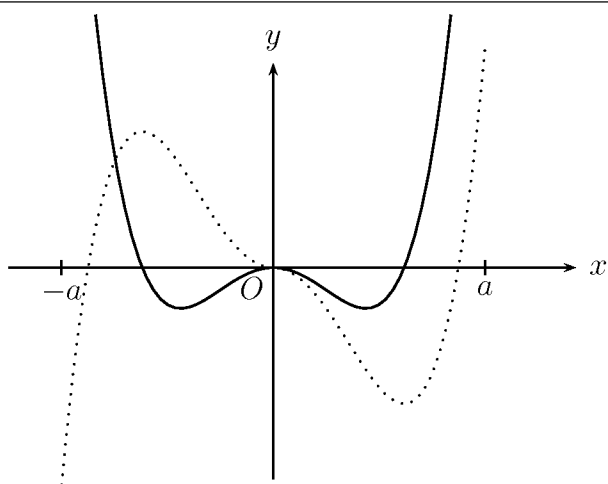
(i) Copy the diagram and continue the graph for  $-a \leq x < 0$ .

**Solution:**



(ii) On another diagram, sketch  $y = f'(x)$ .

**Solution:**



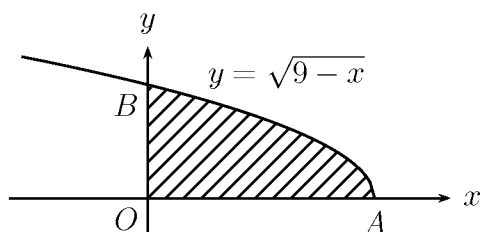
## Section B

(Use a separate writing booklet.)

Marks

### Question 3 (20 marks)

(a)



The shaded area is bounded by the curve  $y = \sqrt{9-x}$  and the co-ordinate axes.

7

(i) Write down the values of  $A$  and  $B$ .

**Solution:**  $A(9, 0)$ ,  $B(0, 3)$

(ii) Find the area  $AOB$ .

**Solution:** 
$$\int_0^9 (9-x)^{\frac{1}{2}} dx = \left[ -1 \times \frac{2}{3} \times (9-x)^{\frac{3}{2}} \right]_0^9,$$
$$= -\frac{2}{3}\{0 - 27\},$$
$$= 18.$$

(iii) Find the volume when the shaded area is rotated about  
( $\alpha$ ) the  $x$ -axis,

**Solution:** Note that  $y^2 = 9 - x$ .

$$\begin{aligned} \text{Volume} &= \pi \int_0^9 (9-x) dx, \\ &= \pi \left[ 9x - \frac{x^2}{2} \right]_0^9, \\ &= \pi \left\{ 81 - \frac{81}{2} - 0 \right\}, \\ &= \frac{81\pi}{2}. \end{aligned}$$

( $\beta$ ) the  $y$ -axis.

**Solution:** Note:  $y^2 = 9 - x$ ,

$$x = 9 - y^2,$$

$$x^2 = 81 - 18y^2 + y^4.$$

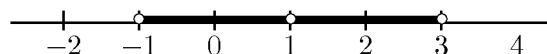
$$\begin{aligned} \text{So Volume} &= \pi \int_0^3 (81 - 18y^2 + y^4) dy, \\ &= \pi \left[ 81y - 6y^3 + \frac{y^5}{5} \right]_0^3, \\ &= \pi \left\{ 243 - 162 + \frac{243}{5} \right\}, \\ &= \frac{648\pi}{5}. \end{aligned}$$

(b) Solve the following inequalities and graph the solutions on separate number lines.

4

(i)  $\frac{2}{|x-1|} > 1,$

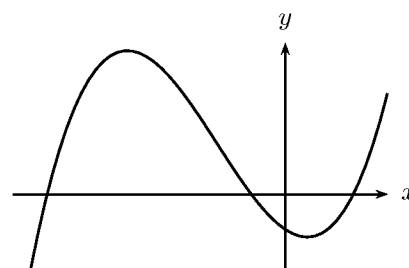
**Solution:**  $2 > |x-1|,$   
 $-2 < x-1 < 2,$   
 $-1 < x < 3, x \neq 1.$



(ii)  $\frac{3}{x-2} \leq \frac{2}{x+1}.$

**Solution:**  $\frac{3}{x-2} - \frac{2}{x+1} \leq 0,$   
 $\frac{3(x+1) - 2(x-2)}{(x-2)(x+1)} \leq 0,$   
 $\frac{x+7}{(x-2)(x+1)} \leq 0,$   
 $(x+7)(x-2)(x+1) \leq 0.$

We note that  $x \neq -1$  and  $x \neq 2,$   
 $\therefore x \leq -7, -1 < x < 2.$



(c) Given  $\triangle ABC$  with vertices  $A(3, 2), B(5, 6)$  and  $C(7, 4):$

5

(i) Find the co-ordinates of  $E$  and  $F$ , the midpoints of  $AB$  and  $BC$  respectively.

**Solution:**  $E$  is  $\left(\frac{3+5}{2}, \frac{2+6}{2}\right) = (4, 4),$   
 $F$  is  $\left(\frac{5+7}{2}, \frac{6+4}{2}\right) = (6, 5).$

(ii) Show that  $EF$  is  
 $(\alpha)$  half of

**Solution:**  $AC^2 = (3-5)^2 + (2-6)^2,$   
 $= 20,$   
 $AC = 2\sqrt{5}.$   
 $EF^2 = (4-6)^2 + (4-5)^2,$   
 $= 5,$   
 $EF = \sqrt{5} = \frac{1}{2}AC.$

$(\beta)$  and parallel to  $AC.$

**Solution:** Slope,  $m_{AC} = \frac{2-4}{3-5} = \frac{1}{2}.$   
 $m_{EF} = \frac{4-5}{4-6} = \frac{1}{2}. \quad \text{i.e., } EF \parallel AC.$

(d) Prove the identity

2

$$\frac{\sin 2A}{1 + \cos 2A} = \tan A.$$

**Solution:**

$$\begin{aligned}\text{L.H.S.} &= \frac{2 \sin A \cos A}{1 + 2 \cos^2 A - 1}, \\ &= \frac{\cancel{2} \sin A \cancel{\cos A}}{\cancel{2} \cos^2 A}, \\ &= \frac{\sin A}{\cos A}, \\ &= \tan A, \\ &= \text{R.H.S.}\end{aligned}$$

(e) Solve  $4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 1$  for  $0 \leq \theta \leq 2\pi$ .  
[HINT: use a double angle result.]

2

**Solution:**

$$\begin{aligned}4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} &= 2 \sin \theta. \\ \text{So } \sin \theta &= \frac{1}{2}, \\ \theta &= \frac{\pi}{6}, \frac{5\pi}{6}.\end{aligned}$$

## Question 4 (14 marks)

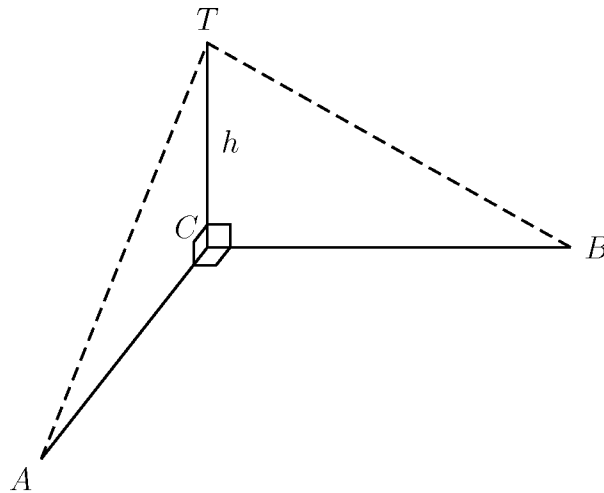
- (a) The graph of  $y = f(x)$  passes through  $(2, 1)$  and  $f'(x) = \frac{x^3 - 4}{x^2}$ . Find  $f(x)$ . 2

**Solution:**

$$\begin{aligned} \int \frac{x^3 - 4}{x^2} dx &= \int (x - 4x^{-2}) dx, \\ &= \frac{x^2}{2} - \frac{4x^{-1}}{-1} + c, \\ &= \frac{x^2}{2} + \frac{4}{x} + c. \end{aligned}$$

So at  $(2, 1)$  we have  $1 = \frac{2^2}{2} + \frac{4}{2} + c$ ,  
*i.e.*,  $c = -3$ .  
Hence  $f(x) = \frac{x^2}{2} + \frac{4}{x} - 3$ .

- (b) 4



$CT$  is a tower of height  $h$ . From  $A$ , due south of the base of the tower, the angle of elevation of  $T$ , the top of the tower, is  $25^\circ$ . From  $B$ , due east, the angle of elevation of  $T$  is  $30^\circ$ . Given that  $AB$  is 200 m, find  $h$  to the nearest metre.

**Solution:**

$$\begin{aligned} \frac{h}{AC} &= \tan 25^\circ \implies AC = h \cot 25^\circ, \\ \frac{h}{BC} &= \tan 30^\circ \implies BC = h \cot 30^\circ. \end{aligned}$$

$$\begin{aligned} AC^2 + BC^2 &= 200^2, \\ h^2 \cot^2 25^\circ + h^2 \cot^2 30^\circ &= 40\,000, \\ h^2 &= \frac{40\,000}{\cot^2 25^\circ + \cot^2 30^\circ}, \\ &\approx 5263.912898. \\ \therefore h &\approx 73 \text{ m (nearest metre).} \end{aligned}$$



(c) Let  $f(x) = \frac{2x}{(1+x^2)^2}$ .

(i) Copy this table and complete it in decimal form.

|        |   |   |   |
|--------|---|---|---|
| $x$    | 1 | 2 | 3 |
| $f(x)$ |   |   |   |

**Solution:**

|        |     |      |      |
|--------|-----|------|------|
| $x$    | 1   | 2    | 3    |
| $f(x)$ | 0.5 | 0.16 | 0.06 |

(ii) Use the table to find values for  $\int_1^3 f(x) dx$  using

( $\alpha$ ) the Trapezoidal rule,

**Solution:** 
$$\int_1^3 f(x) dx \approx \frac{1}{2} \{0.5 + 2 \times 0.16 + 0.06\},$$
  

$$\approx 0.44.$$

( $\beta$ ) Simpson's rule.

**Solution:** 
$$\int_1^3 f(x) dx \approx \frac{1}{3} \{0.5 + 4 \times 0.16 + 0.06\},$$
  

$$\approx 0.4.$$

(iii) Find  $\frac{d}{dx} \left( \frac{1}{x^2 + 1} \right)$ .

**Solution:** 
$$\frac{d}{dx} ((x^2 + 1)^{-1}) = -1 \times 2x(x^2 + 1)^{-2},$$
  

$$= \frac{-2x}{(1 + x^2)^2}.$$

(iv) Using (iii), find the exact value of the integral in (ii).

**Solution:** 
$$\int_1^3 f(x) dx = - \int_1^3 \frac{-2x}{(1 + x^2)^2} dx,$$
  

$$= - \left[ \frac{1}{x^2 + 1} \right]_1^3,$$
  

$$= - \left\{ \frac{1}{10} - \frac{1}{2} \right\},$$
  

$$= 0.4.$$

**End of Paper**

